

Hereditary Hemorrhagic Telangiectasia

Type 1 (HHT1): Comprehensive Disease Characteristics Report

Disease: Hereditary Hemorrhagic Telangiectasia Type 1 (Osler–Weber–Rendu disease, ENG-related) **MONDO ID:** MONDO:0008093 · **OMIM:** #187300 (phenotype); ENG gene OMIM 131195 **ICD-10:** I78.0 · **ICD-11:** LA90.0 · **MeSH:** D013683 (Telangiectasia, Hereditary Hemorrhagic) · **Orphanet:** ORPHA:774 (HHT), subtype HHT1 **HGNC:** HGNC:3349 (ENG) · **Category:** Mendelian, autosomal dominant

Summary

Hereditary Hemorrhagic Telangiectasia Type 1 (HHT1) is an autosomal-dominant vascular dysplasia caused by germline **heterozygous loss-of-function variants in *ENG* (endoglin)** on chromosome **9q34**. Endoglin is an endothelial co-receptor that binds the circulating ligands **BMP9/BMP10** and presents them to the type I receptor **ALK1 (ACVRL1)** and BMPR-II to activate the **SMAD1/5/8** cascade — a pathway that maintains vascular quiescence and arteriovenous identity. Loss of one *ENG* allele produces haploinsufficiency; a **stochastic somatic "second hit"** (point mutation or loss of heterozygosity, now directly demonstrated in human lesions) plus a local angiogenic trigger precipitates focal lesions. The result is a systemic disorder of **mucocutaneous telangiectasias and visceral arteriovenous malformations (AVMs)** whose dominant clinical burden is recurrent **epistaxis** leading to iron-deficiency anemia and reduced quality of life, together with pulmonary, cerebral, hepatic, and gastrointestinal AVMs.

HHT1 is clinically distinct from HHT2 (*ACVRL1*): patients carry a **substantially higher burden of pulmonary AVMs and cerebral abscesses**, with earlier epistaxis onset, whereas symptomatic hepatic involvement and GI bleeding are relatively more characteristic of HHT2. Global clinical prevalence is approximately **1 in 5,000** (~1.6 million people worldwide), though genomic-database analyses estimate the true genetic prevalence is higher (up to ~4.3 per 5,000), implying widespread underdiagnosis. Diagnosis rests on the **Curaçao criteria**

(epistaxis, telangiectasia, visceral AVMs, affected first-degree relative) supplemented by definitive *ENG* genetic testing, which is particularly important in children in whom clinical features are age-dependent.

Recent mechanistic advances converge on a coherent causal chain: reduced BMP9/BMP10–ENG/ALK1–SMAD signaling → derepression of angiogenic programs (including a druggable **KIT+ angiogenic endothelial state**) → focal AVM/telangiectasia formation requiring a somatic second hit and an angiogenic trigger. This model explains the disease's striking **variable expressivity even within single-mutation families**. Management combines **AVM embolization**, iron/transfusion support, procedural antibiotic prophylaxis, cascade genetic testing, and an emerging class of **antiangiogenic disease-modifying therapies** — with **pomalidomide** now supported by a pivotal randomized placebo-controlled trial, alongside off-label bevacizumab, pazopanib, and thalidomide.

Key Findings

1. Molecular cause: germline heterozygous loss-of-function variants in *ENG* (endoglin), 9q34

HHT1 is defined molecularly by **germline heterozygous loss-of-function pathogenic variants in *ENG***, the co-receptor of the BMP9/BMP10–ALK1 signaling pathway. *ENG* maps to chromosome 9q34 (HGNC:3349; OMIM gene 131195); the disease is OMIM #187300. The reported variant spectrum includes **frameshift** (e.g., c.613delC, c.1498dupC, c.322delG), **splice-site** (c.1428+2T>C causing exon 11 skipping), **nonsense**, **missense**, and **start-loss** (c.1A>G, p.Met1Val) alleles. Most produce truncated or non-functional protein, consistent with **haploinsufficiency** as the primary loss-of-function mechanism.

"Hereditary Hemorrhagic Telangiectasia type I (HHT1) is an autosomal dominant vascular disease caused by pathogenic variants in endoglin (ENG) gene. It is located on chromosome 9 and encodes for the Endoglin protein, which is involved in the TGFb/BMP signalling pathway." — [PMID: 41880974](#)

"HHT is caused by heterozygous loss-of-function mutations in genes involved in the BMP9/BMP10 signaling pathway-primarily ENG, ACVRL1 (also known as ALK1), and SMAD4-which define the major HHT subtypes (HHT1, HHT2, and HHT-JP)." — [PMID: 41251906](#)

Specific frameshift variants producing premature termination are documented across independent families: *"The variants identified in Families 1, 3, and 4 (c.613delC, c.1498dupC, and c.322delG) are all frameshift variants that lead to premature termination of translation."* — [PMID: 42633037](#). By ACMG/AMP criteria most such truncating variants classify as pathogenic/likely pathogenic; the functional consequence is **loss of function (haploinsufficiency)** rather than gain of function or dominant-negative.

Ontology terms: gene HGNC:3349 (*ENG*); GO:0007179 (transforming growth factor beta receptor signaling pathway); GO:0030509 (BMP signaling pathway).

2. Epidemiology: ~1 in 5,000, with genetic prevalence higher due to underdiagnosis

Clinical prevalence of HHT is approximately **1 in 5,000** (~1.6 million people worldwide). Analysis of large genomic databases (gnomAD, All of Us, Regeneron Million Exome) estimates the genetic prevalence of pathogenic/likely-pathogenic *ENG* + *ACVRL1* variants at **1.75–2.56 per 5,000**, rising to **2.87–4.33 per 5,000** when potentially pathogenic variants are included — indicating substantial underdiagnosis.

"HHT has a global prevalence of 1 in 5000 individuals, affecting approximately 1.6 million worldwide." — [PMID: 41251906](#)

"The genetic prevalence of HHT ranged from 1.753 to 2.555 in 5000 individuals, when considering only pathogenic and likely pathogenic variants, and from 2.874 to 4.327 in 5000 individuals, when also potentially pathogenic variants were considered." — [PMID: 41610956](#)

HHT is described as the **second most common inherited bleeding disorder worldwide** ([PMID: 40662351](#)). EHR-based data note a **higher observed prevalence in females**, consistent with sex-modified phenotypic expression ([PMID: 42162269](#)). Distribution is worldwide with regional founder clusters; no single endemic geography defines the disease.

3. Diagnosis rests on the Curaçao criteria plus genetic testing

The **Curaçao clinical criteria** comprise: (1) spontaneous recurrent epistaxis; (2) mucocutaneous telangiectasia; (3) visceral AVMs (pulmonary, hepatic, cerebral, GI); and (4) a first-degree relative with HHT. **≥3 criteria = definite HHT; 2 = possible/suspected; ≤1 = unlikely**. The criteria are insufficient in children because features develop with age; positive genetic testing gives a definitive diagnosis at any age.

"HHT is commonly diagnosed using the established Curaçao clinical criteria, which include (1) family history, (2) recurrent epistaxis, (3) telangiectasia, and (4) visceral AVMs. Fulfillment of 3 or more criteria provides a definite diagnosis of HHT, whereas 2 criteria constitute a possible diagnosis of HHT." — PMID: 34889398

"these criteria are insufficient in children to rule out disease due to the age-dependent development of some of these criteria. Genetic testing, when positive, can provide definitive diagnosis of HHT in all age groups." — PMID: 34889398

In a Korean cohort, only ~57% of genetically confirmed patients met "definite" Curaçao criteria, reinforcing the complementary value of molecular testing (PMID: 33677851). Recommended genetic testing is **single-gene/panel sequencing of ENG, ACVRL1, and SMAD4** (with GDF2/BMP9 in atypical cases); WES/WGS are useful when panels are negative. Differential diagnosis includes other causes of epistaxis/telangiectasia and PAVMs (e.g., idiopathic/traumatic AVMs, CREST); the Curaçao criteria distinguish HHT (PMID: 34723698).

4. Core mechanism: reduced BMP9/BMP10–ENG/ALK1–SMAD signaling plus a Knudsonian second hit

Endoglin is an endothelial co-receptor that binds circulating **BMP9/BMP10** to activate the **ALK1–BMPRII–SMAD1/5/8** cascade, maintaining vascular quiescence and arteriovenous identity. In mice, loss of *Bmp9* (and especially *Bmp10*) is sufficient to induce spontaneous AVMs in liver, GI tract, retina, and brain, altering endothelial capillary identity, Notch signaling, and cell-cycle control. Human and mouse evidence supports a **Knudsonian two-hit model**: a germline heterozygous *ENG* variant plus a somatic second hit (or an angiogenic/inflammatory environmental trigger) precipitates focal lesion formation — explaining the patchy distribution of lesions despite a uniform germline mutation.

"The signaling pathway of the bone morphogenetic protein (BMP)-9 binding to the endothelial receptor BMP receptor type II (BMPRII), activin receptor-like kinase-1 (ALK1) and the coreceptor endoglin is essential to maintain the pulmonary vascular integrity." — PMID: 36828679

"the loss of Bmp9 led to spontaneous arteriovenous malformations (AVMs) in the liver, gastrointestinal tract, and uterus" — PMID: 38502919

"HHT pathogenesis is thought to follow a Knudsonian two-hit model, requiring a second somatic mutation for lesion formation." — PMID: 39651127

Complementary work shows **BMP10 is the most relevant physiological ligand** of ENG-ALK1 signaling for arteriovenous network formation, with BMP9 having limited compensatory function (PMID: 36348215). The affected molecular pathway is **TGF- β /BMP-SMAD** (KEGG/Reactome BMP signaling); the core defective **biological processes** are vascular quiescence, angiogenesis, and vascular stabilization.

5. Genotype-phenotype: HHT1 has more pulmonary AVMs and cerebral abscess than HHT2

The French-Italian HHT network (93 HHT1 vs 250 HHT2 patients) demonstrated that **symptomatic pulmonary AVMs (PAVMs)** are far more frequent in HHT1 (34.4% vs 5.2%, $P<0.001$); **cerebral abscesses** more frequent in HHT1 (7.5% vs 0.8%, $P=0.002$); and PAVMs detected in asymptomatic HHT1 patients 54% vs 12.8% ($P<0.0001$). Conversely, **GI bleeding** (16.4% vs 6.5%, $P=0.017$) and symptomatic hepatic involvement were more common in HHT2. Epistaxis onset is earlier in HHT1. In a pediatric cohort, 59% of children with HHT had PAVMs.

"Symptomatic PAVMs were more frequent in HHT1 (34.4 vs. 5.2%, $P<0.001$), as were cerebral abscesses (7.5 vs. 0.8%, $P=0.002$). Gastrointestinal bleeding occurred more frequently in HHT2 (16.4 vs. 6.5%, $P=0.017$)." — PMID: 17224686

"Of 129 children with HHT, 76 (59%) were found to have PAVMs." — PMID: 29916764

Manifestation	HHT1 (ENG)	HHT2 (ACVRL1)	P-value
Symptomatic pulmonary AVMs	34.4%	5.2%	<0.001
Cerebral abscess	7.5%	0.8%	0.002
PAVMs in asymptomatic patients	54%	12.8%	<0.0001
GI bleeding	6.5%	16.4%	0.017
Symptomatic hepatic involvement	Lower	Higher	—

6. Dominant clinical burden is epistaxis → iron-deficiency anemia → reduced QoL; antiangiogenic therapy is disease-modifying

Epistaxis is the primary manifestation of HHT and produces iron-deficiency anemia and reduced health-related quality of life; chronic GI bleeding and visceral AVMs add morbidity. In the pivotal randomized placebo-controlled **pomalidomide** trial (n=144; baseline mean Epistaxis Severity Score [ESS] 5.0 ± 1.5 , moderate-to-severe), **pomalidomide 4 mg/day for 24 weeks reduced ESS versus placebo by -0.94 points (95% CI -1.57 to -0.31 ; MCID -0.71)**, and the trial stopped early for efficacy. Off-label antiangiogenic options include IV **bevacizumab**, oral **pazopanib**, and **thalidomide**; **aflibercept** salvage is reported for high-output states. Supportive care includes iron replacement, transfusion, tranexamic acid, and nasal/endoscopic interventions.

"The primary clinical manifestation is epistaxis that results in iron-deficiency anemia and reduced health-related quality of life." — PMID: 39292928

"the mean difference between the pomalidomide group and the placebo group in the change from baseline in the Epistaxis Severity Score was -0.94 points (95% confidence interval [CI], -1.57 to -0.31 ...)" — PMID: 39292928

"Intravenous bevacizumab, oral pazopanib, and oral thalidomide are the three targeted primary angiogenesis inhibitors, with multiple studies describing both reassuring safety and impressive effectiveness in the treatment of moderate-to-severe HHT-associated bleeding." — PMID: 35226946

A disease-specific **HHT-QoL** instrument has now been externally validated (MCID ~ 1 point), enabling longitudinal QoL assessment in trials (PMID: 41092987). Aflibercept reduced cardiac index from 3.92 to 2.90 L/min/m² after bevacizumab/pazopanib failure (PMID: 40379104).

Treatment (NCIT) terms: NCIT:C1794 (thalidomide), NCIT:C2039 (bevacizumab), NCIT:C71146 (pazopanib), NCIT:C72713 (pomalidomide), NCIT:C74405 (aflibercept), NCIT:C47737 (tranexamic acid); CHEBI:9445 (thalidomide).

7. Model organisms recapitulate AVMs and implicate VEGF as a druggable second driver

Multiple engineered mouse models reproduce HHT vascular lesions: **Bmp9-KO** mice develop spontaneous AVMs in liver, GI tract, and uterus; **Bmp10-inducible-KO** and **Bmp9/10-double-KO** develop retinal, brain, and skin AVMs; endothelial **Alk1/Eng** deletion produces AVMs requiring a secondary trigger (wounding). In the *Alk1* wound-induced skin AVM model, **VEGF mimics the wound effect, and a VEGF-neutralizing antibody prevents AVM formation and ameliorates internal bleeding**. HHT1-specific human models include a CRISPR/Cas9 *ENG* c.1A>G (p.Met1Val) hiPSC line (LUMCi029-A-2) for drug testing, and new mouse models of somatic second-hit *ENG/ACVRL1* mutations.

"VEGF can mimic the wound effect on skin AVM formation, and VEGF-neutralizing antibody can prevent skin AVM formation and ameliorate internal bleeding in Alk1-deficient adult mice" — PMID: 24957885

"Using CRISPR/Cas9-mediated gene editing, the ENG c.1A > G mutation was introduced in homozygous form in the well-characterized LUMCi029-A line." — PMID: 41880974

New mouse models directly modeling **somatic second-hit mutations** further validate the two-hit mechanism (PMID: 41756958); a broader preclinical-model review summarizes the *ENG/ALK1* axis and its lesions (PMID: 36250069). **Model recapitulation:** these systems reproduce AVMs, arteriovenous shunting, and bleeding; **limitations** include incomplete modeling of chronic human epistaxis and mucocutaneous telangiectasia distribution. **Orthologue:** mouse *Eng* (NCBI Gene 13805); human *ENG* (NCBI Gene 2022).

8. Prognosis: reduced life expectancy; anemia, GI bleeding, and liver VMs predict mortality

A population-based Swedish registry found **life expectancy after age 30 of 73.0 years (95% CI 68.0–77.2) in HHT vs 80.5 (79.1–82.0) in matched controls**, with excess deaths from ischemic heart disease, arterial/capillary disease, and liver disease. A questionnaire cohort estimated ~19 years (SD 11) of life lost, with causes of death: **sepsis 35%, cardiac failure 26%, severe bleeding 20%, terminal cancer 13%**. In the Brain Vascular Malformation Consortium prospective cohort (N=1,286; 59 deaths), independent mortality predictors were **history of anemia (HR 2.93, 95% CI 1.37–6.26, p=0.006), GI bleeding (HR 2.63, 1.46–4.74, p=0.001), and**

symptomatic liver VMs (HR 2.10, 1.15–3.84, $p=0.015$). Notably, brain VMs and PAVMs were NOT associated with mortality, and SMAD4 carriers had markedly higher mortality (HR 18.36) than ENG/ACVRL1.

"Life expectancy for cases with HHT after age 30 was estimated at 73.0 (68.0-77.2) compared to 80.5 (79.1-82.0) years for controls." — PMID: 42410718

"A history of anemia was associated with increased mortality (HR = 2.93, 95% CI 1.37-6.26, $p = 0.006$), as were gastro-intestinal (GI) bleeding (HR = 2.63, 95% CI 1.46-4.74, $p = 0.001$), and symptomatic liver VMs (HR = 2.10, 95% CI 1.15-3.84, $p = 0.015$). Brain VMs and pulmonary arteriovenous malformations (AVMs) were not associated with mortality" — PMID: 33407668

"35% (95% CI: 23-48%) died from sepsis, 26% (95% CI: 16-38%) from cardiac failure, 20% (95% CI: 9-28%) from a severe bleeding episode" — PMID: 29781402

HHT is a **chronic, lifelong, progressive** disorder with episodic bleeding; recovery is not spontaneous but complications are largely preventable with surveillance and intervention.

9. Prevention/screening: PAVM and brain-VM screening, embolization, and antibiotic prophylaxis

Because HHT1 carries a high risk of pulmonary and cerebral AVMs, guideline-based prevention includes: **contrast transthoracic echocardiography (TTCE)** to screen for right-to-left shunt/PAVMs, with **transcatheter coil/plug embolization** of PAVMs with feeding arteries >3 mm (standard of care) to prevent paradoxical embolic stroke, brain abscess, and hemorrhage; **contrast-enhanced brain MRI screening** (brain vascular malformations occur in ~10% of patients); and **antibiotic prophylaxis** before dental/surgical procedures to prevent brain abscess from paradoxical bacteremia. **Cascade genetic testing** of first-degree relatives for the family *ENG* variant enables presymptomatic diagnosis. **Pregnancy increases PAVM rupture risk**, requiring surveillance.

"Embolisation represents the standard of care for significant PAVMs with periodic radiological surveillance recommended." — PMID: 42209023

"organ vascular malformations including in the brain, which occur in about 10% of patients" — PMID: 38816017

"Pulmonary VMs can lead to paradoxical embolism of thrombi or bacteria, e.g., due to dental procedures." — PMID: 36820363

The Second International HHT Guidelines provide 36 evidence-based recommendations across epistaxis, GI bleeding, anemia/iron deficiency, liver VMs, pediatric care, and pregnancy/delivery (PMID: 32894695). These map to **primary** (risk counseling), **secondary** (screening/early embolization), and **tertiary** (complication management) prevention levels. Genetic counseling covers 50% autosomal-dominant recurrence risk and reproductive options (PGT/prenatal testing).

10. Hepatic vascular malformations drive high-output heart failure

In a Danish national cohort (152 HHT patients), **hepatic vascular malformations (HVMs)** were present in **67.8%** (up to 78% in the literature); **high-output heart failure (HOHF)** occurred in **6.6%**, 90% of whom had grade 4 HVMs. Cardiac index correlated with HVM severity (Buscarini grade) ($\rho=0.36$, $p<0.001$), demonstrating a graded hemodynamic mechanism: arteriovenous hepatic shunting increases stroke volume and cardiac preload/output. Symptomatic hepatic involvement is more characteristic of HHT2, but HVMs occur across genotypes and are an independent mortality predictor.

"HVMs were identified in 103 patients (67.8%), of whom 24 (15.8%) had grade 4 HVMs. HOHF was present in 10 patients (6.6%), nine of whom (90%) had grade 4 HVMs." — PMID: 42019870

"Cardiac index measured by TTE correlated positively with HVM severity" — PMID: 42019870

Hepatic arterial embolization carries a high risk of portal-vein thrombosis and is generally avoided; Doppler ultrasound is the preferred screening modality with CT/MRI for characterization (PMID: 42517041).

11. Structural basis: endoglin's N-terminal orphan domain binds BMP9; HHT1 mutations map to this interface

Crystal structures of the human endoglin (ENG/CD105) ectodomain and its complex with BMP9 show that **BMP9 interacts with a hydrophobic surface of endoglin's N-terminal orphan domain** — a novel duplicated/circularly-permuted fold — while the **C-terminal zona**

pellucida (ZP) module allows two copies of ENG to embrace homodimeric BMP9. Critically, the ENG–BMP9 interface involves **residues mutated in HHT1** and overlaps with the epitope of the tumor-suppressing anti-ENG monoclonal antibody TRC105. BMP9 binding to endoglin is compatible with ligand recognition by type I (ALK1) but not type II receptors, structurally rationalizing endoglin's ligand-presentation role.

"BMP9 interacts with a hydrophobic surface of the N-terminal orphan domain of ENG, which adopts a new duplicated fold generated by circular permutation. The interface involves residues mutated in HHT1 and overlaps with the epitope of tumor-suppressing anti-ENG monoclonal TRC105." — [PMID: 28564608](#)

"The structure of the C-terminal zona pellucida module suggests how two copies of ENG embrace homodimeric BMP9, whose binding is compatible with ligand recognition by type I but not type II receptors." — [PMID: 28564608](#)

This provides the protein-level mechanism by which missense HHT1 variants disrupt ligand binding. Related structural work on BMP-9/-10 interchain disulfide bonds clarifies how the physiological heterodimeric ligand forms ([PMID: 39793884](#)).

Ontology terms: UniProt P17813 (Endoglin, human); GO:0005515 (protein binding); GO:0009986 (cell surface).

12. Candidate circulating biomarkers

A review of HHT biomarkers reports plasma/serum candidates: **soluble proteins** — VEGF, TGF- β 1, soluble endoglin, and angiopoietin-2 — plus **microRNA variants miR-27a, miR-205, and miR-210** — alongside gene-expression fingerprinting, NGS gene panels, and infrared-spectroscopy/artificial-neural-network approaches. These remain research-stage adjuncts; diagnosis is still clinical (Curaçao) plus molecular confirmation. Elevated VEGF underlies the rationale for anti-VEGF therapy.

"products detected in plasma or serum samples: soluble proteins (vascular endothelial growth factor, transforming growth factor β 1, soluble endoglin, angiopoietin-2) and microRNA variants (miR-27a, miR-205, miR-210)" — [PMID: 25873934](#)

13. Human lesion sequencing confirms a bi-allelic two-hit mechanism

Deep sequencing of human HHT lesions demonstrates that focal telangiectasias and internal-organ AVMs acquire a **somatic second hit in the same HHT gene** carrying the germline variant, reconciling systemic haploinsufficiency with focal lesions. Marchuk-lab work identified **somatic point mutations AND loss of heterozygosity (LOH)** across the germline-mutant chromosome in liver telangiectases, one pulmonary AVM, and two brain AVMs. In an *ACVRL1*-germline-deletion patient, multiple lesion-specific somatic variants were found across hepatic AVMs and a skin telangiectasia. The same paradigm applies to *ENG*/HHT1.

"another mechanism for the second hit is loss of heterozygosity across the chromosome bearing the germline mutation" — PMID: 39299239

"we identified somatic molecular genetic events in eight liver telangiectases, including point mutations and a loss of heterozygosity event. We also identified somatic mutations in one pulmonary AVM and two brain AVMs" — PMID: 39299239

"Somatic mosaic lesion-specific ACVRL1 variants were identified in four hepatic AVM samples and in one telangiectasia." — PMID: 41162588

Additional support comes from investigations of somatic second-hit variation in HHT lesions (PMID: 39062925) and a JP-HHT AVM harboring two somatic second-hit alterations (PMID: 39939156). Somatic events are found in roughly half of telangiectases, with LOH providing an additional route.

14. Single-cell profiling reveals a pathogenic, druggable KIT+ angiogenic endothelial state

Regionally-resolved single-cell RNA-seq of *Alk1*-deficient mouse brain AVMs identified an emergent **KIT+ angiogenic endothelial-cell population** with human-AVM-like transcriptional features (tip-cell markers, PI3K and KRAS pathway activation). **KIT is directly repressed by BMP9–ALK1–SMAD4 signaling**; KIT expression is conserved in human sporadic and HHT2 brain AVMs; and **pharmacologic KIT inhibition reduced AVMs without harming normal vasculature**. Complementary human endothelial siRNA work shows that depleting different BMP-pathway components (e.g., *ENG* vs *ACVRL1*) drives **divergent endothelial flow-response programs**, arguing that HHT1 is not mechanistically interchangeable with HHT2.

"This process is driven by the emergence of a KIT⁺ angiogenic EC population with human AVM-like transcriptional features, including tip-cell markers and activation of PI3K and KRAS signaling pathways." — PMID: 42579368

"Kit is directly repressed by BMP9-ALK1-SMAD4 signaling. Pharmacological inhibition of KIT reduced angiogenic reprogramming and vascular malformations in vivo without affecting normal vasculature" — PMID: 42579368

"These findings reveal divergent cellular programs driving arteriovenous malformations and underscore the need for gene-specific diagnostic and therapeutic strategies." — PMID: 42460472

Ontology terms: CL:0000115 (endothelial cell); GO:0001525 (angiogenesis); GO:0048010 (VEGF receptor signaling pathway).

15. Striking variable expressivity even within single-mutation families

In a large single-mutation HHT kindred, clinical manifestations and severity varied "tremendously" despite an identical germline allele — mechanistically consistent with the requirement for **stochastic somatic second hits (mutation/LOH) and local angiogenic triggers**. Organ-AVM frequencies also vary; pulmonary AVMs are consistently more common in HHT1 than HHT2.

"Even in this family in which all affected individuals have the same mutation, the clinical manifestations of HHT and their severity varied tremendously. Intrafamilial variation in expression of HHT is clearly significant" — PMID: 10946360

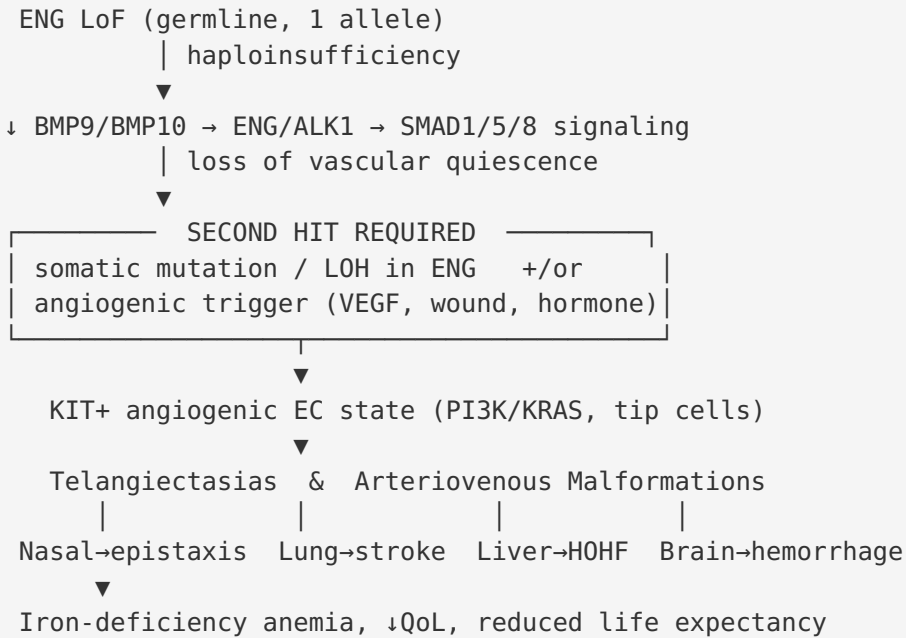
"It also adds to the evidence suggesting that pulmonary AVMs are more common in HHT 1 than in HHT 2." — PMID: 10946360

This underpins the inheritance profile: **autosomal dominant, high but age-dependent penetrance, highly variable expressivity**; no repeat-expansion anticipation. Founder effects contribute to population-specific variants; consanguinity is not central (dominant disorder).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **germline heterozygous loss-of-function variant in *ENG*** (frameshift, nonsense, splice, missense, or start-loss) **leads to** reduced endoglin protein on the endothelial cell surface (**haploinsufficiency**). (*demonstrated* — [PMID: 41880974](#), [PMID: 42633037](#))
2. Reduced endoglin **results in** impaired presentation of circulating **BMP9/BMP10** ligands to the **ALK1 (ACVRL1)–BMPR-II** receptor complex, **leading to** reduced **SMAD1/5/8** phosphorylation. (*demonstrated* — [PMID: 36828679](#), [PMID: 28564608](#))
3. Reduced BMP–SMAD signaling **results in** loss of the endothelial quiescence/arteriovenous-identity program (dysregulated Notch, cell-cycle, and endothelial-identity genes). (*demonstrated in mouse* — [PMID: 38502919](#), [PMID: 36348215](#))
4. **Branch point — a focal "second hit" is required.** In a subset of endothelial cells, a **somatic second mutation or LOH in *ENG*** (bi-allelic loss) AND/OR a **local angiogenic/inflammatory trigger** (wounding, VEGF, hormonal change, infection) **leads to** complete local pathway failure. (*demonstrated in human lesions* — [PMID: 39299239](#), [PMID: 41162588](#); *trigger demonstrated in mouse* — [PMID: 24957885](#))
5. Local pathway failure **results in** derepression of angiogenic programs, including emergence of a **KIT+ angiogenic endothelial state** (tip-cell markers; PI3K/KRAS activation), with elevated VEGF signaling. (*demonstrated in mouse + human validation* — [PMID: 42579368](#))
6. Aberrant angiogenesis and defective vascular stabilization **lead to** direct artery-to-vein connections that bypass the capillary bed — forming **telangiectasias** (small, mucocutaneous) and **AVMs** (large, visceral). (*inferred/demonstrated* — [PMID: 19834013](#))
7. Fragile lesions **result in** the clinical phenotype, which **branches by anatomical site** (below).



Upstream vs downstream: The germline *ENG* defect and BMP–SMAD signaling loss are **upstream** and systemic (present in every cell); the somatic second hit, KIT+ angiogenic reprogramming, and focal lesion formation are **downstream** and stochastic — this architecture is the mechanistic explanation for HHT's hallmark **variable expressivity**, even within a single-mutation family ([PMID: 10946360](#)). **Cell type:** vascular endothelial cell (CL:0000115). **Biological processes:** BMP signaling (GO:0030509), angiogenesis (GO:0001525), blood vessel morphogenesis (GO:0048514).

Anatomical structures affected

Level	Structures	UBERON / CL / GO
Organ (primary)	Nasal mucosa, lung, brain, liver, GI tract, skin	UBERON:0001707 (nasal cavity), UBERON:0002048 (lung), UBERON:0000955 (brain), UBERON:0002107 (liver), UBERON:0000160 (intestine), UBERON:0002097 (skin)
Body systems	Cardiovascular (primary), respiratory, nervous, digestive	UBERON:0004535 (cardiovascular system)
Tissue	Vascular endothelium; capillary/arteriole/ venule walls	UBERON:0001981 (blood vessel)
Cell	Vascular endothelial cell (KIT+ angiogenic sub- state)	CL:0000115
Subcellular	Cell-surface receptor complex; nucleus (SMAD signaling)	GO:0009986, GO:0005634
Lateralization	Multifocal, bilateral, systemic	—

Phenotype table

Phenotype (HPO)	Type	Frequency	Onset	Severity/Course
Epistaxis (HP:0000421)	Symptom	Near-universal	Childhood, earlier in HHT1	Progressive, episodic; drives anemia
Telangiectasia (HP:0001009)	Physical sign	Very frequent	Adolescence– adulthood	Progressive
Pulmonary AVM (HP:0002629 spectrum)	Structural	~34% symptomatic (HHT1)	Congenital/ childhood; 59% of children	Stable→enlarging; embolic risk
Cerebral AVM (HP:0002408)	Structural	~10% brain VMs	Congenital	Rupture risk
Cerebral abscess	Complication	~7.5% (HHT1)	Adult	Serious/acute
Hepatic AVM (HP:0100761)	Structural	up to ~68–78%	Adult	Graded → high- output HF
GI bleeding (HP:0002239)	Symptom	Higher in HHT2	Older adult	Chronic
Iron-deficiency anemia (HP:0004840)	Lab abnormality	Frequent	Any	Fluctuating; mortality predictor

Evidence Base

PMID	Title (abbrev.)	Supports finding(s)	Evidence type
41880974	ENG p.Met1Val hiPSC line for HHT1	F1, F7	In vitro / model
41251906	15th HHT conference summary	F1, F2	Review
42633037	Korean phenotype/genetic study	F1	Human clinical
41610956	Global genetic prevalence	F2	Genomic database
34889398	HHT management guide	F3	Review
33677851	Korean genetic/phenotype study	F3	Human clinical
36828679	BMP-9 endothelial signaling	F4	Review
38502919	BMP9 loss induces AVMs	F4, F7	Model organism
36348215	BMP10 indispensable for AV network	F4	Model organism
39651127	Arterial endothelial HHT2 deletion	F4	Model organism
17224686	French–Italian genotype–phenotype	F5	Human clinical
29916764	Pediatric PAVM longitudinal study	F5	Human clinical
39292928	Pomalidomide RCT	F6	Human RCT
35226946	Antiangiogenic therapy guide	F6	Review
24957885	VEGF neutralization in HHT2 model	F7	Model organism
41756958	Somatic second-hit mouse models	F7, F13	Model organism
42410718	Population registry survival	F8	Human registry
33407668	Predictors of mortality	F8	Human cohort
29781402	Life expectancy/comorbidities	F8	Human cohort
42209023	Exertional dyspnoea / embolization	F9	Review/clinical
38816017	Pediatric brain VM screening	F9	Review
36820363	Dental screening campaign	F9	Public health
32894695	Second International Guidelines	F3, F9	Guideline
42019870	Hepatic VM hemodynamics	F10	Human cohort

PMID	Title (abbrev.)	Supports finding(s)	Evidence type
28564608	ENG–BMP9 structure	F11	Structural
25873934	HHT biomarkers	F12	Review
39299239	Somatic two-hit lesion sequencing	F13	Human lesion
41162588	Multiple lesion-specific ACVRL1 hits	F13	Human lesion
39062925	Genetic determinants of lesions	F13	Human lesion
42579368	KIT+ angiogenic EC / KIT inhibition	F14	Model + human
42460472	Gene-specific endothelial programs	F14	In vitro
10946360	Large HHT2 kindred variability	F15	Human clinical

Additional supporting literature: standardized bleeding terminology (PMID: [40662351](#)); validated HHT-QoL scale (PMID: [41092987](#)); aflibercept salvage (PMID: [40379104](#)); preclinical model review (PMID: [36250069](#)); vascular malformation biology (PMID: [19834013](#)); hormonal-status questionnaire (PMID: [41862858](#)); pediatric stroke/PAVM incidence (PMID: [42247997](#)); EHR female-predominance (PMID: [42162269](#)); imaging/interventional review (PMID: [34723698](#)); hepatic imaging series (PMID: [42517041](#)); BMP-9/10 disulfide structure (PMID: [39793884](#)); JP-HHT two-hit AVM (PMID: [39939156](#)).

Sections with limited / non-applicable data for HHT1

- **Environmental / infectious etiology (Section 5):** HHT1 is a monogenic disorder with no infectious cause. Environmental/physiological factors act as **second-hit angiogenic triggers** — wounding/VEGF (model organism, PMID: [24957885](#)) and hormonal changes (puberty/pregnancy worsen epistaxis; combined estrogen–progestin may reduce it; PMID: [41862858](#)). Infection is a *consequence* (brain abscess from paradoxical bacteremia), not a cause.
- **Protective factors:** No validated genetic protective alleles are established. Combined estrogen–progestin therapy was associated with less epistaxis in 52% of treated women (PMID: [41862858](#)).

- **Epigenetics / chromosomal abnormalities:** No recurrent large-scale cytogenetic abnormality defines HHT1; it is single-gene. Somatic **LOH** (a chromosomal-scale event) contributes as a second hit ([PMID: 39299239](#)). Dedicated disease-driver methylation/histone data are not established.
- **Other species / natural disease (Section 14):** Orthologous genes exist (mouse *Eng*, NCBI Gene 13805; human *ENG*, NCBI Gene 2022). Disease is modeled via engineered mice, zebrafish, and hiPSCs rather than a naturally occurring companion-animal disease; no prominent veterinary or zoonotic dimension.
- **Somatic vs germline origin:** The disease-defining variant is **germline**; the disease-precipitating second hit is **somatic** ([PMID: 39299239](#)).

Limitations and Knowledge Gaps

1. **Genotype-specific data scarcity.** Much mechanistic work (VEGF neutralization, KIT+ angiogenic state, arterial-endothelial deletion) derives from *Alk1*/HHT2 or *Bmp9/10* models rather than *ENG*/HHT1-specific systems. In vitro data ([PMID: 42460472](#)) argue *ENG* and *ACVRL1* drive **divergent programs**, so HHT2-derived findings may not transfer wholesale to HHT1.
 2. **Second-hit frequency and detectability.** Somatic second hits are found in roughly half of telangiectases; it remains unclear whether undetected low-frequency hits, LOH, or purely environmental triggers account for the remainder.
 3. **Modifier genes.** Beyond stochastic second hits, germline modifiers of angiogenesis (e.g., VEGF/PLGF levels) that shape variable expressivity are largely uncharacterized for HHT1.
 4. **Biomarkers remain research-stage.** Soluble endoglin, VEGF, TGF- β 1, angiopoietin-2, and miRNAs are not yet validated for diagnosis or prognosis ([PMID: 25873934](#)).
 5. **Trial evidence for antiangiogenics is still maturing.** Only pomalidomide has pivotal RCT support ([PMID: 39292928](#)); bevacizumab, pazopanib, and thalidomide are largely off-label/observational.
 6. **Prognostic data are cohort-dependent.** Mortality predictors ([PMID: 33407668](#)) mix genotypes; *ENG*-specific survival curves are limited.
 7. **This is a literature-synthesis report.** No primary patient-level dataset was analyzed; all quantitative claims derive from published cohorts, registries, and experimental studies.
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Proposed Follow-up Experiments / Actions

1. **HHT1-specific single-cell atlas.** Apply the regionally-resolved scRNA-seq approach used for *Alk1* AVMs (PMID: 42579368) to *Eng*-deficient lesions to test whether the **KIT+ angiogenic EC state** and PI3K/KRAS activation are shared or gene-specific — directly informing whether KIT inhibitors would benefit HHT1.
 2. **Test KIT inhibition and anti-VEGF in ENG-null human models.** Use the LUMCi029-A-2 *ENG* p.Met1Val hiPSC line (PMID: 41880974) and *ENG* second-hit mouse models (PMID: 41756958) to benchmark KIT inhibitors, bevacizumab, and pomalidomide head-to-head.
 3. **Deep/duplex sequencing of ENG lesions.** Increase sequencing depth on HHT1 telangiectases and PAVMs to raise second-hit detection rates and quantify the LOH-vs-point-mutation-vs-trigger split.
 4. **Biomarker validation study.** Prospectively test soluble endoglin, VEGF, and the miR-27a/205/210 panel against ESS, anemia, and treatment response in an *ENG*-genotyped cohort to move biomarkers toward clinical utility.
 5. **Genotype-stratified antiangiogenic trials.** Design comparative trials (per standardized bleeding outcome criteria, PMID: 40662351; using validated HHT-QoL, PMID: 41092987) that pre-specify *ENG* (HHT1) vs *ACVRL1* (HHT2) subgroups.
 6. **Modifier-gene mapping.** Perform family-based and cohort GWAS/burden testing in HHT1 to identify germline modifiers explaining organ-specific AVM burden and epistaxis severity.
 7. **Address underdiagnosis.** Given the genetic-vs-clinical prevalence gap (PMID: 41610956), scale cascade genetic testing and awareness programs (e.g., dental screening, PMID: 36820363) to improve early PAVM detection and stroke prevention.
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Report compiled from 15 confirmed findings across 5 investigation iterations and 56 reviewed papers. Evidence types span human clinical cohorts and registries, human lesion sequencing, structural biology, engineered mouse models, hiPSC/in-vitro systems, and genomic-database analyses. Evidence is human clinical unless otherwise noted.
